Nitin B S

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Final-year Computer Science Engineering student at VVCE, Mysuru, focused on backend engineering — distributed pipelines and applied AI.

Education

- Vidyavardhaka College of Engineering, Mysuru (B.E. in CS) | 2023 – 2027 (Expected) — CGPA: 9.09
- Sri Jayachamarajendra PU College, Mysuru (P.U.) — Percentage: 89.5%
- Acharya Vidya Kula, Mysuru (Secondary Education, ICSE) — Percentage: 89.1%

Skills

- **Languages:** C, C++, SQL | **Familiar with**: Python, JavaScript
- **AI / ML:** LangChain, Hugging Face Transformers, LoRA fine-tuning, RAG
- **DevOps & Tools:** Git, Docker, Kubernetes, AI-assisted development (Claude Code, Antigravity)
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering

Experience

HPE Career Preview Program (CPP) | Search Engine for Object Storage System — **Feb 2026 – Jul 2026**

- Built a hybrid-search RAG pipeline over MinIO object storage, combining dense vector (kNN) and BM25 keyword retrieval in OpenSearch, with Redis caching, containerised with Docker and deployed on Minikube.
- Engineered an event-driven ingestion pipeline: MinIO upload events stream through Kafka to Python workers, Apache Tika extracts text and metadata, and content is chunked and embedded before indexing into OpenSearch, with a Dead Letter Queue isolating failed documents.
- Enabled cross-modal search using Jina CLIP v2, embedding both text and images into a single 512-dim vector space so one query retrieves matching documents and images from a single index.

Projects

3D Warehouse Visualization and Vacancy Optimization System

- Built an Odoo 19 Community add-on that turns a warehouse rack layout into real Inventory storage locations, generated automatically. Changing a layout updates those locations without deleting any, and is refused if it would remove a slot that still holds stock.
- Built an interactive 3D view of the warehouse inside Odoo (OWL + Three.js) that shows how full each slot is from live inventory data. It updates in real time over Odoo's message bus as receipts and deliveries are processed, with slot search, a first-person walk-through mode, and a 2D drag-and-drop layout editor.
- Created Pluggable storage-optimization strategies that extend Odoo's putaway logic: 3D bin-packing (FFDH with centre-of-mass, BLF with weight stratification), feeding a storage-location assignment stage (W-BFDH) and two picking-efficiency strategies: golden-zone and S-shape routing.

LLM-Based Log Parser using LoRA Fine-Tuning

- Fine-tuning Qwen3-4B to convert unstructured system logs into structured event templates and parameters, replacing hand-written regex rules that break whenever log formats change.
- Used LoRA (rank 16, Hugging Face PEFT) to train small low-rank adapter weights on a frozen base model, making fine-tuning feasible on a single T4 GPU instead of updating all 4B parameters.
- Prepared public benchmark log datasets into instruction-style training pairs, with held-out log sources for testing so the model is evaluated on unseen formats rather than memorised templates.

Extracurricular Activities

- Volunteer Teacher | U&I (NGO)
- Peer Tutor
- Coordinator | Programming Workshop by Department